Araz Mehrabani

Mechanical Engineer | Control · Mechatronics · Robotics

22880 Wedel / Hamburg
+49 1575 436 3238
araz.h.mehrabani@gmail.com
Date of birth: 28.07.1989

PORTFOLIO
Technical
Project Examples
Google Drive

PROFILE & CORE COMPETENCIES

Mechanical engineer with research and industrial experience in **control systems, mechatronics and dynamic simulation**. Focus on 6-DOF system modeling, MATLAB/Simulink, PID dynamic positioning, actuator-fault analysis, electromechanical mechanisms and prototyping.

TECHNICAL FOCUS

Dynamics & Control: 6-DOF equations of motion, MATLAB/Simulink, closed-loop control, PID/Dynamic Positioning, translational/rotational and velocity control.

Controller Evaluation: iterative PID tuning; overshoot, settling time, steady-state error, tracking error; complete failure of 1 or 2 of 4 actuators and time-delay scenarios.

Mechatronics: mechanisms, motor/actuator selection, torque/speed/ratio, wheel/rail loads, mechanical-electrical integration and prototype testing.

Embedded & Prototyping: Arduino, Marlin firmware, G-code, Slic3r, STL, stepper calibration, hardware wiring; ABS/PLA, layer height, infill, bed/nozzle temperature and print speed.

PROFESSIONAL EXPERIENCE

Development Engineer — Sabaniroo, Tehran **01/2022–02/2023**

- Developed a two-axis **PV-panel cleaning system**: rail/wheel guidance plus belt-driven carriage, torque/speed/ratio and load calculations, motor selection, CAD and prototype testing; approx. **10% lower material use**.

Mechanical Engineer — TehranRigzar, Tehran **03/2017–12/2021**

- Developed and dynamically designed vibration and conveying equipment; converted a vibrating screen to direct-excitation vibration motors and performed static/modal ANSYS analyses plus experimental testing of spring elements.

Student Research Assistant — Hakim Sabzevari University, Sabzevar **12/2014–12/2016**

- Modeled a hovering AUV in **MATLAB/Simulink**, developed a PID-based Dynamic Positioning controller for precise underwater welding tasks, and investigated time delays and actuator failures; results were published and presented in multiple technical papers/conferences.

Working Student – Mechanical Design — Ario Fidar Maham Co., Tehran **11/2012–05/2013**

- Recreated a 40x90 jaw crusher in **CATIA V5**, including individual components, assemblies and manufacturing drawings.

SELECTED DEVELOPMENT

Low-Cost Hand Prosthesis – Embedded Prototyping — modular prosthesis prototype; rebuilt a 3D printer using Arduino/Marlin, firmware/G-code changes, stepper calibration, hardware wiring and Slic3r process settings.

EDUCATION

M.Sc. Wind Energy Engineering, Hochschule Flensburg — current grade: 1.9 **03/2023–11/2026**
M.Sc. Mechanical Engineering, Hakim Sabzevari University — Control & Robotics **09/2014–01/2017**
B.Sc. Mechanical Engineering, Islamic Azad University — Structural Dynamics **09/2008–06/2013**

LANGUAGES

German: B2 (actively improving). English: C1 / IELTS 7.0. Persian: Native.